

Models Performance Report

1. Logistic Regression Model Performance Interpretation

Classification Report:

- The model correctly predicts Sideways trend (43% of the time) followed by Uptrend (37%) and Downtrend (35%) but overall, the model prediction precision is not better, below 50%. (Precision)
- The model is better in identifying the actual Uptrend (82%) worse for actual Downtrend (15%) and Sideways (8%). The model is poorly recalling or missing to identify Downtrend and Sideways. (Recall)
- The model detects best Uptrend (51%, the strongest class) but struggles to detect Sideways (0.13) and Downtrend (0.21) which are the weakest classes. (F1-scores)
- Overall, the model gets 37% of predictions correct (accuracy of 37%)

Confusion Matrix:

- The model heavily favors predicting “Uptrend,” even when the true label is “Downtrend” or “Sideways.” Uptrend dominates predictions, and Downtrend and Sideways are mostly misclassified.
- The model is predominantly predicting Uptrend (1386 out of 1696) and heavily misclassifying Downtrend (1,189 out of 1,495 Downtrend cases predicted as Uptrend) and Sideways (1,158 out of 1,454 cases predicted as Uptrend).

Summary:

Uptrend is the best predicted class in the Logistic Regression model because:

- Uptrend is the best predicted class as it is correctly identified in over 81% of actual cases, has the best balance of precision and recall among all classes (F1-score for Uptrend: 0.51059), low Precision for Uptrend (37%) but still better than Downtrend. And misclassifications are relatively low compared to other classes.

In Contrast with Other Classes

- 85% of actual Downtrends cases are missed (Recall: 0.15) and most are misclassified as Uptrend.
- 92% of actual Sideways cases are missed (Recall: 0.08) and heavily misclassified as Uptrend.

The Logistic Regression model is strongest at identifying Uptrend, but it's biased toward Uptrend predictions, which leads to poor performance on Downtrend and Sideways.

2. Random Forest Model Performance Interpretation

Classification Report Interpretation

- **Uptrend** is the best-predicted class ($F1 = 0.68$), with strong precision(67%) and recall(68%).
- **Downtrend** is reasonably well predicted ($F1 = 0.64$).
- **Sideways** is the weakest ($F1 = 0.48$), suggesting the model struggles to distinguish it.

Confusion Matrix Interpretation

- The model is strongest at identifying Uptrends, with 68% recall and 1156 correct predictions.
- Sideways trends are ambiguous with nearly half misclassified either as Downtrend or Uptrend.
- Downtrend predictions are solid, but some confusion with Sideways exists.

Summary

The Random Forest model performs well overall (60% accuracy), especially for Uptrend and Downtrend while Sideways remains a challenge !

The model limitations:

- Sideways class ambiguity: May reflect overlapping feature patterns or lack of distinctive signals.
- Class imbalance: Uptrend has slightly more support, which may bias the model.
- Volatility sensitivity: Outliers may distort feature scaling and decision boundaries.
- No macro context: The model lacks awareness of external events (e.g., earnings, news).